Applied Deep Learning

WiSe 2025/26

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Before ChatGPT

All sorts of implementation work possible

- Research papers without code
- Code in the wrong language (e.g., Tensorflow instead Pytorch)
- …

Idea of the course:

Learn programming with a DL language and implement/translate existing ideas

Examples from 2021/22

- Translate Vprop/Vadam implemented in PyTorch into TensorFlow
- Implement Top-KAST in PyTorch
- DNN Mode connectivity in TensorFlow

UCI_Datasets	fixed print fn	8 months ago
.gitignore	Implemented tensorflow base optimizer class	8 months ago
README.md	add results	8 months ago
Result table.png	add results	8 months ago
experiment.py	Fix errors	8 months ago
main.py	Add all experiments	8 months ago
model.py	Cleanup	8 months ago
requirements.txt	add results	8 months ago
train_and_test.py	Fix errors	8 months ago
v_optimizers.py	add results	8 months ago

README.md

Optimizer VADAM and Vprop in TensorFlow

- Converting VADAM and Vprop Optimizers from PyTorch to TensorFlow
- Adoptation of the optimizers on UCI datasets.
- This project is based on the paper Fast and Scalable Bayesian Deep Learning by Weight-Perturbation in Adam
- In order to get datasets and compute the results (rmse, log likelihood), the repository DropoutUncertaintyExps

November 2022: ChatGPT

Now all the translations can be done via an LLM

- Using most recent versions this works quite reliably
- And/or students do not really learn anything

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⇒ Adapt the course structure

Until last semester

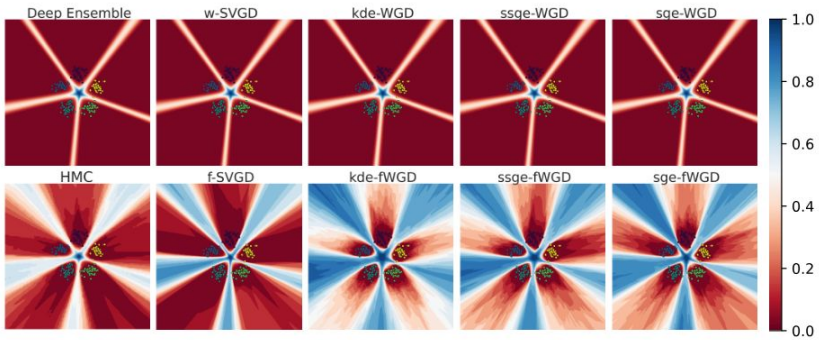
Adapt to ChatGPT era:

- **Idea:** Work on state-of-the-art research and implement a DL-related project
- **Application:** Students applied for a bunch of proposed projects based on ideas developed by the supervisors

Examples from last semester(s)

Background:

Deep ensembles face challenges in maintaining diversity among members, which limits ensemble performance an affects uncertainty estimates. Introducing a kernelized repulsive term in training mitigates this issue.



Learning goals:

- Understand [repulsive deep ensembles]
- Learn to read PyTorch (research) code
- Learn to write JAX code
- Run experiments on uncertainty quantification tasks

Supervisor: Emanuel

Examples from last semester(s)

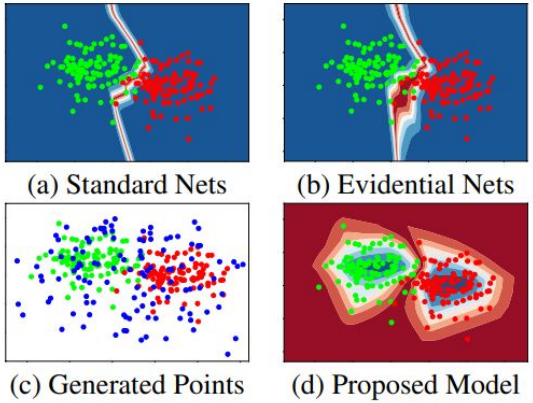
Background: This is implementation of Evidential deep learning as well as variational autoencoders and generative adversarial networks in TF that incorporated to automatically generate out-of-distribution exemplars for training.

code:https://muratsensoy.github.io/gen.html

Learning Goals

- Learn to read and write PyTorch/TF code
- Learn coding on generative models
- Learn about UQ techniques

Supervisor: Mina



source: https://arxiv.org/pdf/2006.04183

Until last semester

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  - Scaling this is not possible while student numbers are growing
  - Many dropouts which makes the initial effort all for nothing
  - The less the topic aligns with our expertise, the harder it is to assess students' own contribution

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Now

1. **Kick-off Meeting**: Presentation of course structure
2. **Project Proposal**: During the semester,
   - ➢ students work in **groups of 2–4**
   - ➢ to prepare a **2-page proposal** outlining their research idea,
   - ➢ including implementation details, and, if meaningful, unit tests, experimental design, and/or comparison to existing approaches.
3. **Project Work**: Students work
   - ➢ at their own pace on the project,
   - ➢ keeping in mind that the expected effort corresponds to the 6 ECTS awarded for the course.
   - ➢ They may consult the supervisor in case of questions.
4. **Submission**: Any time until the end of the semester. Submissions can take the form of a code repository, a developed software package, or a research paper.
5. **Evaluation**: based on proposal's originality, methodological soundness, and quality of implementation.

Your next step

1. Find 1–3 team members
2. Look at recent publications from Mina and David's group (or other people from the department working in DL★)
3. Search the current literature and work out a project idea
4. Write the proposal by laying out the current gap in research and motivate your proposed solution
5. Send the proposal to David + the person you want to work with

★ Before approaching them, talk to us first!